Maya Maciel-Seidman

maya.maciel-seidman@duke.edu | (516) 946-3191

ORCID iD: [0009-0006-1758-0752](https://orcid.org/0009-0006-1758-0752) | GitHub: github.com/mlmac-seid

Education

2030 **PhD in Earth and Climate Sciences, Duke University**
Advisor: Prof. Jonathan C. Ryan

2025 **Bachelor of Arts in Earth and Environmental Sciences, Vanderbilt University**
Honors: *cum laude*
Minors: Data Science; Environmental and Sustainability Studies

Research Appointments

Naval Research Enterprise Internship Program (NREIP) — Washington, D.C.
U.S. Naval Research Laboratory Remote Sensing Division — May 2024–December 2025
Advisor: Dr. Trina Merrick

- Ran CNN models to delineate, categorize, and map ice wedge polygons on the Alaska North Slope.
- Conducted fieldwork in July 2025 in Utqiagvik, AK to ground truth remote sensing data.
- Built machine learning models to predict changes in active layer thickness on the Alaska North Slope.

Undergraduate Researcher — Nashville, TN
Climate, Health, and Energy Equity Lab (CHEEL), Vanderbilt University — January 2023–May 2025
Advisors: Dr. Zdravka Tzankova, Dr. Carol C. Ziegler

- Developed quantitative methodology to calculate carbon offset value of energy efficient retrofits to low-income housing.

Undergraduate Research Assistant — Nashville, TN
Mixed Intelligence Development for Programming Lab, Vanderbilt University — January 2022–June 2022
Advisor: Dr. Yu Huang

- Contributed to study design to determine the effect of ambient environment on programmer performance.

Research Intern — Raleigh, NC
Game2Learn Lab, North Carolina State University — June 2021–August 2021
Advisor: Dr. Tiffany Barnes

- Created and tested computing-infused K-12 curriculum using UC Berkeley's Snap! language.

Awards

2026 **Best Student Oral Presentation –** International Symposium on Artificial Intelligence in Glaciology

2026 **National Science Foundation Graduate Research Fellowship Honorable Mention**

2025 **Vanderbilt University Outstanding Leadership in Earth and Environmental Sciences Award**

2021–2025 **Cornelius Vanderbilt Scholarship –** Vanderbilt University full tuition merit scholarship and funded summer immersive experience

Peer-Reviewed Publications

Maciel-Seidman, M. L., Merrick, T.L., Liang, R., Grossman, S.M., Richards, IV, D.F., Vermillion, M., Abelev, A. (2026) Ice-Wedge Polygon Geomorphometry: A Comparative Assessment Using Deep Learning and Repeat Lidar. *Scientific Reports* (under review)

Richards, IV, D.F., Merrick, T.L., Abelev, A., Vermillion, M., **Maciel-Seidman, M.L.**, Grossman, S.M. (2026) Remote Sensing-Based Framework for Detecting and Interpreting Permafrost Terrain Hydrologic Connectivity. *Frontiers in Earth Science* (accepted)

Karas, Z; Page, D; **Maciel-Seidman, M**; Sharafi, Z; Huang, Y (2026) The Influence of Environmental Odors on Student Programmers During Code Comprehension and Code Writing. *ACM Transactions on Software Engineering and Methodology*, doi.org/10.1145/3822406

Maciel-Seidman M, Tzankova Z, Ziegler CC, Lele A, Lu S, Yan Y and Muchira JM (2024) Mobilizing carbon offsetting to reduce energy cost burdens: a new approach for calculating and monetizing the offset value of energy efficiency upgrades to low-income housing. *Front. Energy Res.* 12:1437560, doi.org/10.3389/fenrg.2024.1437560

Maciel-Seidman, M. L., Channapatna, R., & Channapatna, S. (2020). Soil Moisture Sensing Robot: A Novel Agricultural Device. *Journal of Dawning Research*, 2, 15-33. http://dawningresearch.org/Papers/2020/JDR2020002.pdf

Conference Presentations

2026 **Maciel-Seidman, M.**, & Ryan, J., Modeling Greenland Ice Sheet catchment-scale meltwater runoff using transfer learning, Oral Presentation, *International Symposium on Artificial Intelligence in Glaciology, Hanover, New Hampshire*

2026 Richards IV, D., Merrick, T., Liang, R., Abelev, A., Vermillion, M., **Maciel-Seidman, M.**, and Grossman, S., Remote Sensing-Based Framework for Detecting and Interpreting Permafrost Terrain Hydrologic Connectivity, *EGU General Assembly, Vienna, Austria*

2024 **Maciel-Seidman, M. L.**, Merrick, T. L., Grossman, S. M., Richards, D. F., Abelev, A., Vermillion, M. S., Liang, R. T., Analysis of Active Layer Thickness and Climate Data at Utqiagvik, Alaska with Random Forest and Multiple Linear Regression Algorithms, *AGU Annual Meeting, Washington, DC*

2024 Merrick, T. L., Richards, D. F., Grossman, S. M., **Maciel-Seidman, M. L.**, Abelev, A., Vermillion, M. S., Liang, R. T., Arctic ice wedge polygon characterization using multilevel remote sensing, *AGU Annual Meeting, Washington, DC*

2024 Richards, D.F., Merrick, T. L., Grossman, S. M., **Maciel-Seidman, M. L.**, Abelev, A., Vermillion, M. S., Liang, R. T., Remote Sensing of Periglacial Trough Structure Variability and Connections to Ice-Wedge Degradation, *AGU Annual Meeting, Washington, DC*

2024 Grossman, S. M., Merrick, T. L., Richards, D. F., **Maciel-Seidman, M. L.**, Abelev, A., Vermillion, M. S., Liang, R. T., Multilevel remote sensing investigation of Tundra surface variability in Utqiagvik, Alaska, *AGU Annual Meeting, Washington, DC*

Software and Datasets

Maciel-Seidman, M., Merrick, T., Liang, R., Grossman, S., Richards, IV, D., Vermillion, M., & Abelev, A. (2026). Ice-Wedge Polygon Geomorphometry: A Comparative Assessment Using Deep Learning and Repeat Lidar Surveys. Zenodo. https://doi.org/10.5281/zenodo.19578055

Fieldwork

July 2025 **Utqiagvik, Alaska –** Carried out flight planning and UAS safety checks to enable collection of lidar data and hyperspectral and RGB imagery of ice wedge polygons. Measured soil active layer thickness, logged Trimble GeoXH points, collected vegetation and soil samples, and installed an array of soil moisture sensors and an automatic weather station. Conducted this fieldwork for a U.S. Naval Research Laboratory project to quantify and characterize spatiotemporal changes in Arctic periglacial terrain.

Sept 2024 **Collegiate Peaks, Colorado –** Collected till samples, measured clast sizes, and calculated angle of repose to relatively date moraines from the Bull Lake and Pinedale (Pleistocene) glaciations. Conducted this fieldwork for Vanderbilt University course EES 4440: Glacial Geology.

Aug–Dec 2023 **Tennessee State University Agricultural Research and Education Center –** Augered soil samples for hydrocarbon analysis and created soil profiles. Collected and filtered water samples from research wells to analyze oxygen and hydrogen isotopes, dissolved inorganic carbon, and volatile organic compounds. Conducted this fieldwork for Vanderbilt University course EES 3280: Environmental Geochemistry.

Teaching

Vanderbilt University, Nashville, TN

Spring 2025	Grader, DS 1000, How Data Shape Our World
Fall 2024	Grader, DS 1000, How Data Shape Our World
Spring 2024	Grader, DS 1000, How Data Shape Our World
Fall 2023	Teaching Assistant, ENVS 1101, Foundations of Climate Studies
Fall 2023	Grader, DS 1000, How Data Shape Our World
Spring 2023	Grader, DS 1000, How Data Shape Our World
Fall 2022	Grader, DS 1000, How Data Shape Our World

Professional Memberships and Affiliations

International Glaciological Society (2026–present)
Association of Polar Early Career Scientists (2025–present)
International Association of Cryospheric Sciences (2025–present)
American Geophysical Union (2024–present)
Phi Sigma Rho – Alpha Eta Chapter (2022–present)
National Center for Women & Information Technology (2020–present)

Relevant Courses

Glaciology: Glacial Geology; Field Investigations: Colorado
Computing: Programming and Problem Solving with Python; Program Design and Data Structures; Data Science: How Data Shape Our World; Applied Machine Learning; Fundamentals of Data Science; Agent-Based Modeling; Geospatial Data Science; Environmental Spatial Data Analysis; Software Engineering for Spatial Data Analysis
Wilderness Medicine: National Outdoor Leadership School (NOLS) Wilderness First Aid

Skills

Computing: Python, R, Bash, MATLAB, C++, Java, NetLogo, GNU nano, Vim, Git, GitHub, High-Performance Computing, Slurm, MPI
Geospatial Software: QGIS, ArcGIS Pro, GeoDa, Google Earth Pro, GeoMapApp, QGroundControl
Field equipment/techniques: Hach FH950 velocity flow meter, lightweight deflectometer, Trimble GeoXH, ASD FieldSpec, tile probe, Kestrel 5500 Weather Meter, soil auger, clinometer
CAD and 3D Printing Software: Solid Edge, Autodesk Inventor, Ultimaker Cura, Bambu Studio